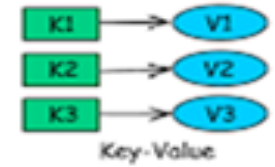




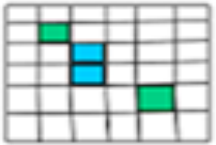
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NoSQL



Key-Value



Column Store



Graph



Document

WEB TECHNOLOGIES

CSE-4004

LECTURE 1: NoSQL DATABASES



By: Dr.MD.Sirajuddin

SCOPE

NoSQL DATABASE

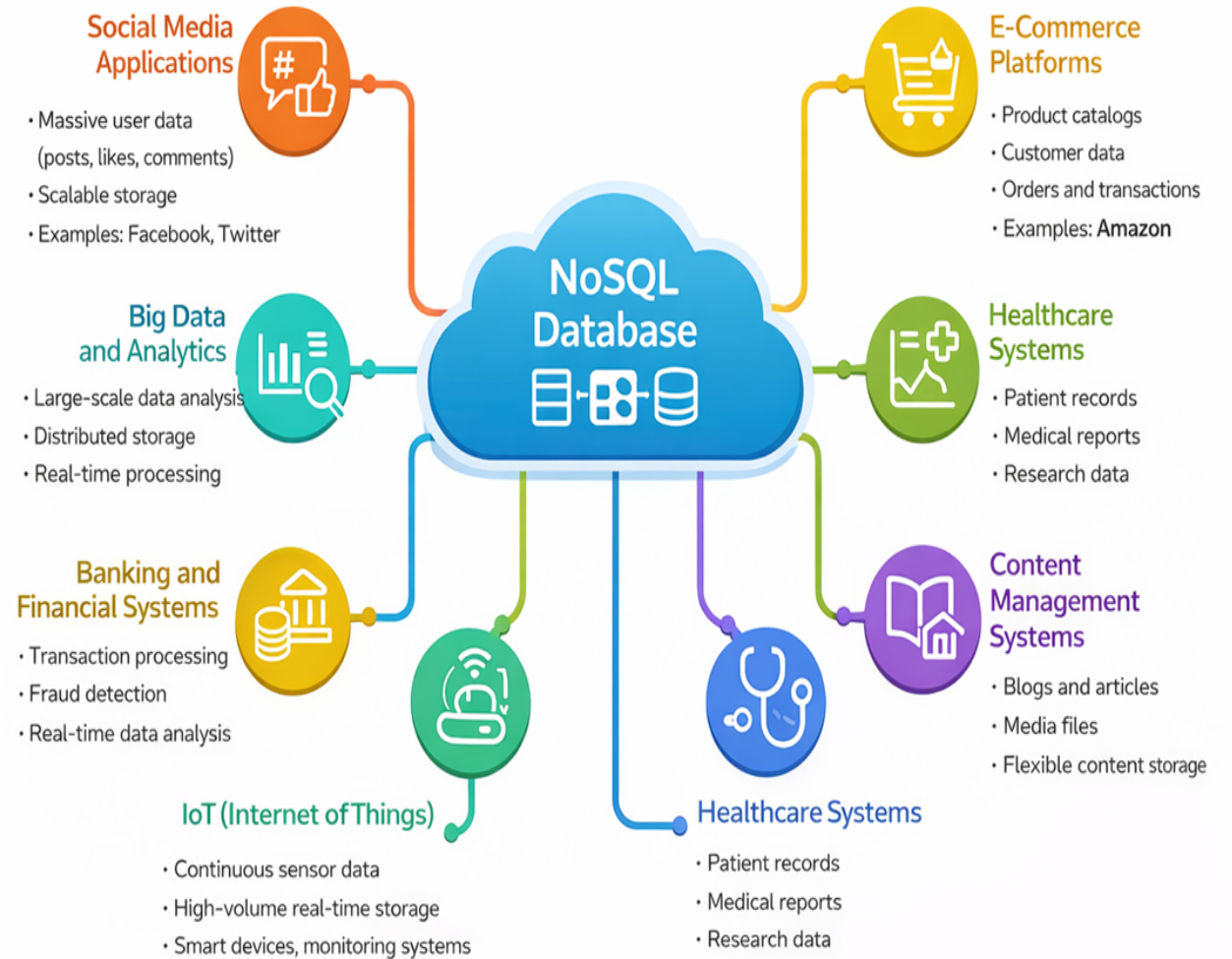
- **NOT**
- **ONLY**
- **SQL**
- NoSQL was first introduced in 1998 by Carlo Strozzi – Open Source Database (Strozzi NoSQL).
- Later, in 2009, the term NoSQL was popularized again by Eric Evans
- At that time, NoSQL became widely known as "Not Only SQL"
- It refers to any non-relational database
- It is designed to handle large amounts of semi-structured or unstructured data
- Provides flexible data models that are highly scalable

APPLICATIONS OF NoSQL DATABASES

➤ NoSQL databases are widely used in applications that require:

- High scalability
- Fast data processing
- Flexible data storage
- Handling of large and unstructured data

Applications of NoSQL Databases



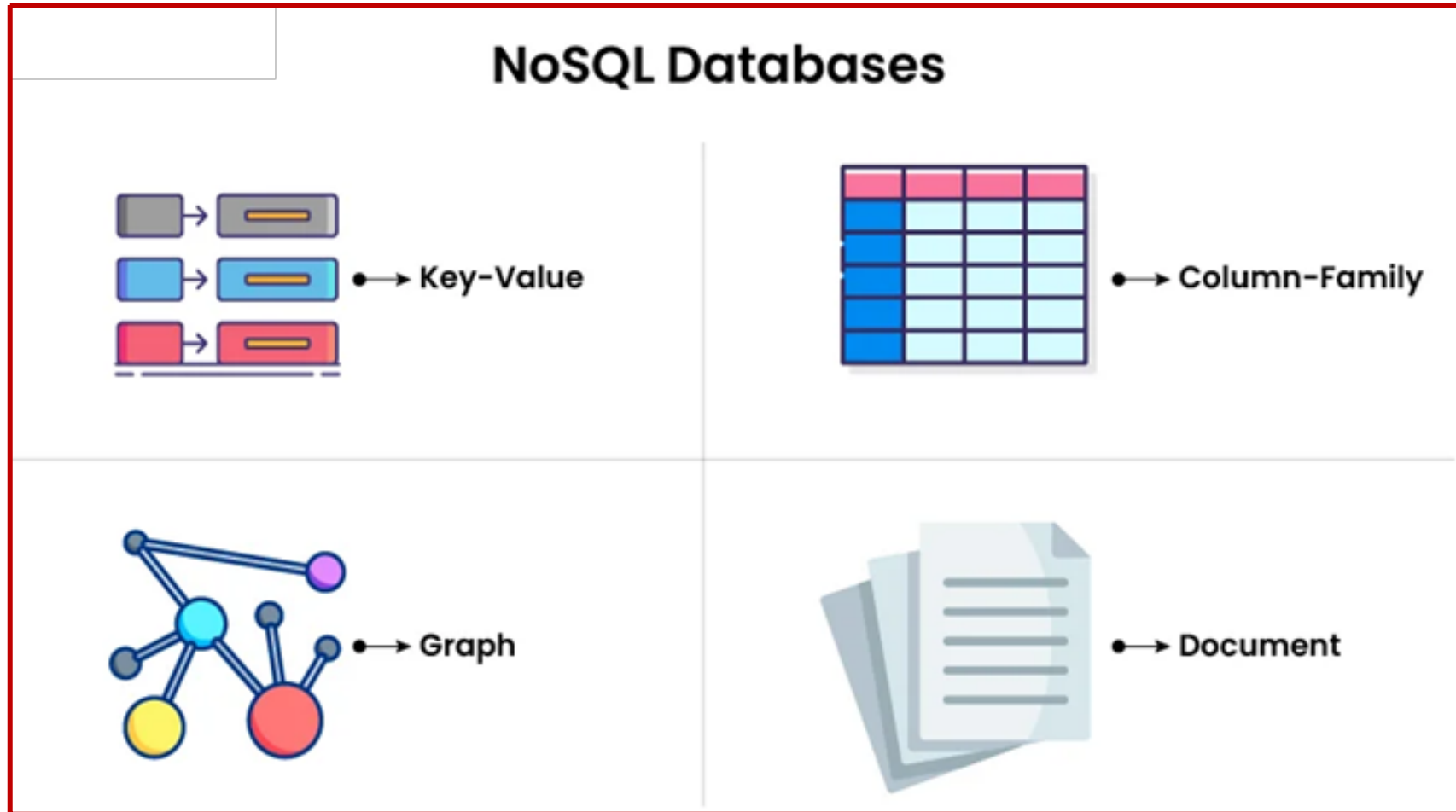
TYPES - NoSQL DATABASE

- **Over time, four major types of NoSQL database have emerged:**

- 1. Document Databases**
- 2. Key-Value Databases**
- 3. Wide-column stores**
- 4. Graph Databases.**

- **Nowadays, multi-model databases are also becoming quite popular.**

NoSQL DATABASE TYPES



DOCUMENT DATABASE

- A document-oriented database stores data in documents similar to **JSON** objects.
- Each document contains pairs of fields and values.
- The values can typically be a **variety of types**, including things like strings, numbers, booleans, arrays, or even other objects.
- A document database offers a flexible data model, much suited for **semi-structured** and typically **unstructured** data sets.
- They also support **nested structures**, making it easy to represent complex relationships or hierarchical data.

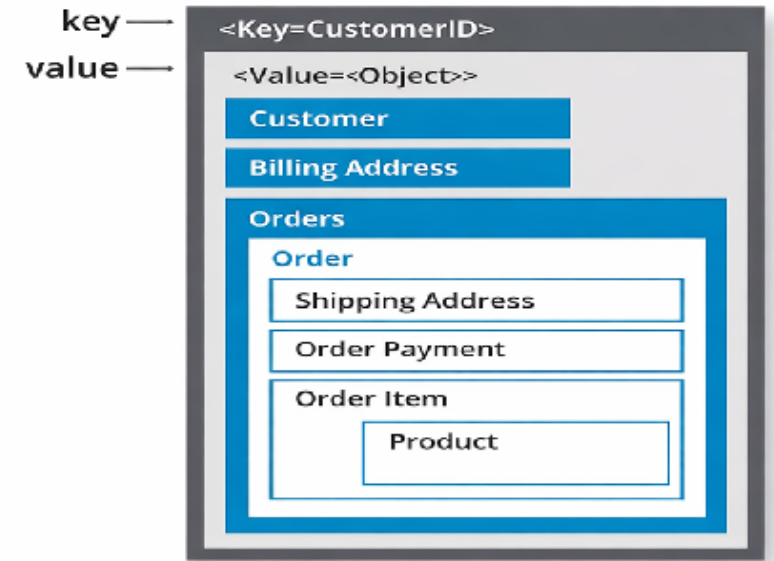
Key	Document
1001	<pre>{ "CustomerID": 99, "OrderItems": [{ "ProductID": 2010, "Quantity": 2, "Cost": 520 }, { "ProductID": 4365, "Quantity": 1, "Cost": 18 }], "OrderDate": "04/01/2017" }</pre>



KEY-VALUE DATABASE

- A key-value store is a simpler type of database where each item contains keys and values.
- Each key is unique and associated with a single value.
- They are used for caching and session management.
- Provide high performance in reads and writes because they tend to store things in memory.

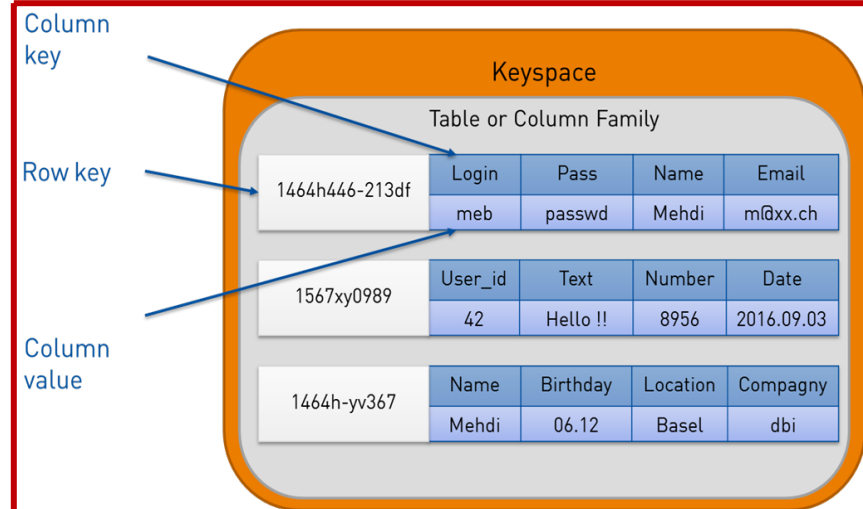
key	value
123	123 Main St.
126	(805) 477-3900



WIDE COLUMN DATABASE

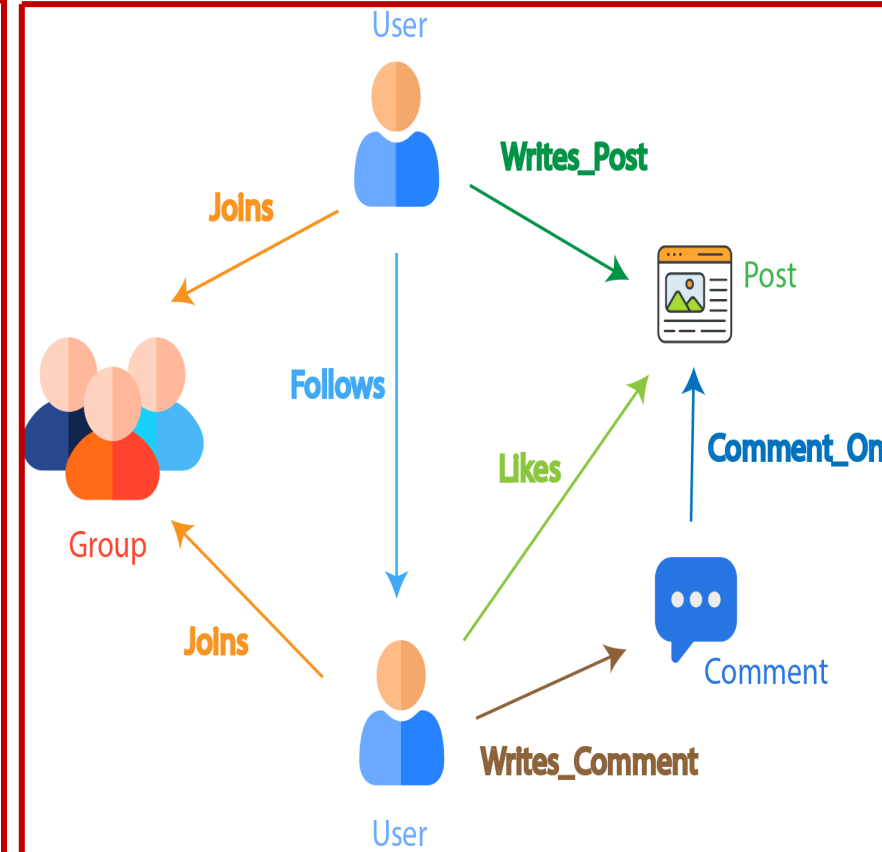
- Each row can have millions of columns
- Columns can vary from row to row
- That is why it is called “**wide**” — a row can contain a very large number of dynamic columns.

Row A	Column 1	Column 2	Column 3
	Value	Value	Value
Row B	Column 1	Column 2	Column 3
	Value	Value	Value



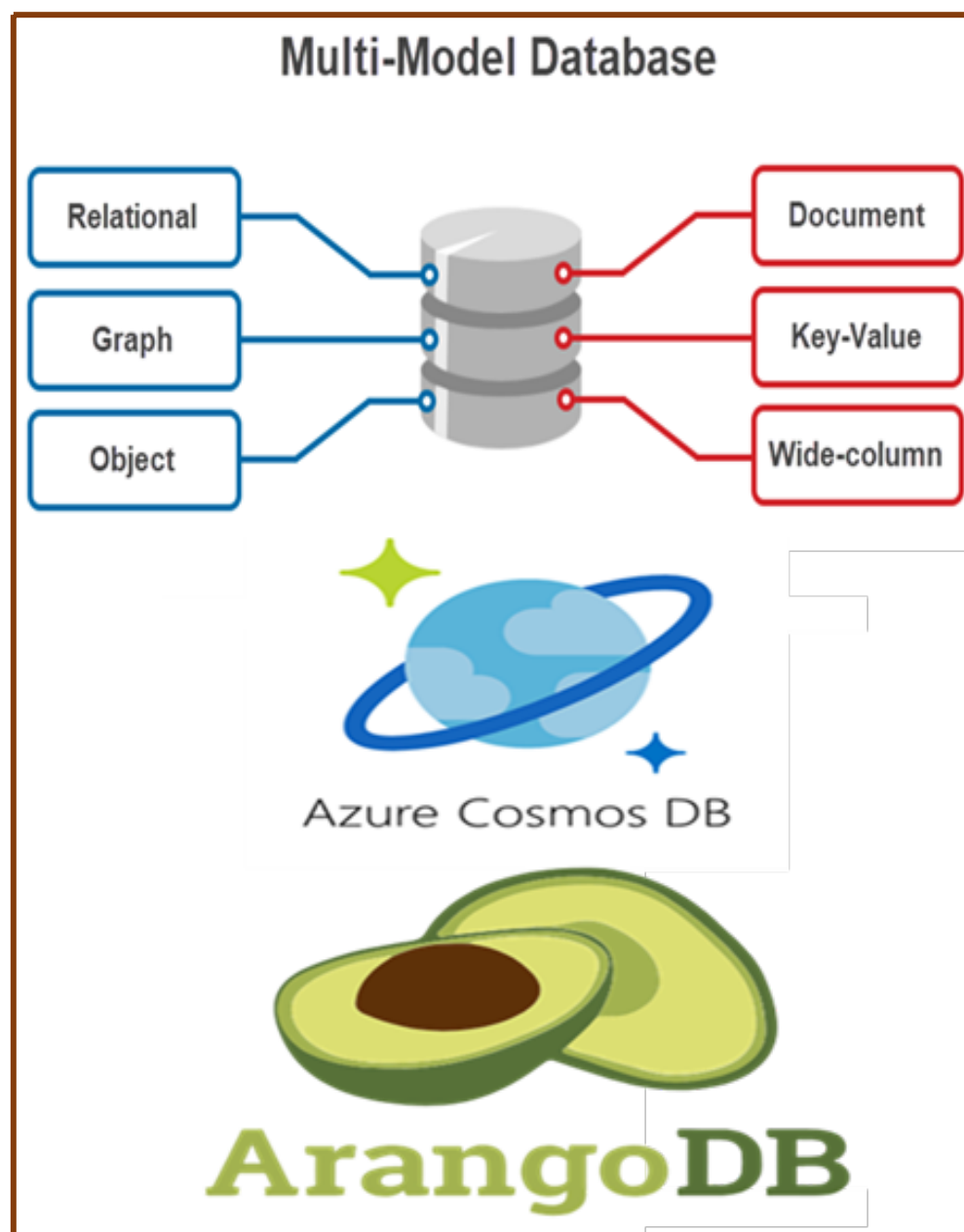
GRAPH DATABASE

- A graph database stores data in the form of nodes and edges.
- Nodes typically store information about people, places, and things), while edges store information about the relationships between the nodes.
- They work well for highly connected data.
- Used to manage complex relationships between data. User → Friends → Friends of Friends (facebook)
- Nodes (entities)
- Edges (relationships)
- Properties (attributes)



MULTI MODEL

- Multi-model databases support more than one type of NoSQL data model so that developers can choose based on their application requirements.
- These databases have a unified database engine that can handle multiple data models within a database instance.



KEY FEATURES & IMPORTANCE OF NoSQL Databases

- **Distributed Computing** → data is stored on multiple servers or in a cluster of nodes → improves availability and reliability
- **Ease of use for Developers**
- **Big Data** → They can store and process huge amounts of structured, semi-structured, and unstructured data
- **High Scalability** → They support **horizontal scaling** (adding more servers, distribute data across them)
- **Flexible Schema** → does not require fixed structure (table format).
- **High Performance** → Fast data access for real-time applications.
- **Suitable for Modern Applications** → Used in social media, IoT, e-commerce, analytics, etc.

Data Types

Data Type	Example
String	"SCOPE"
Integer	21
Float	99.5
Boolean	true
Date	2026-02-19
Array	["Math", "Science"]
Object / Document	{"city": "Delhi"}
Null	null

Document

- A document is a record stored in JSON format.

```
{  
  "id": 1,  
  "name": "Siraj",  
  "age": 21,  
  "city": "Hyderabad"  
}
```

- Each document is independent
- Structure can vary
- Stored inside a collection

Nested Document

- A document inside another document.

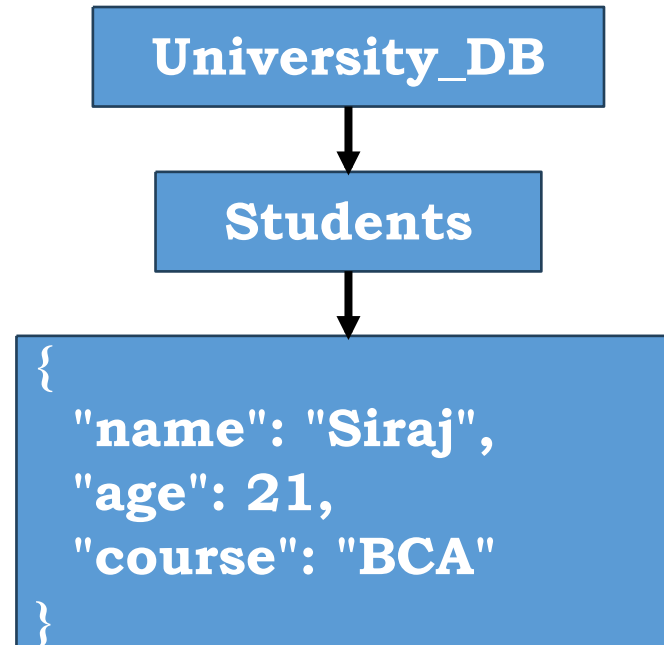
```
{  
  "name": "Bob",  
  "age": 21,  
  "address": {  
    "city": "Guntur",  
    "state": "Andhra Pradesh",  
    "pincode": 522001  
  }  
}
```

Importance

- Store related data together
- Faster reads
- Reduces joins
- Better performance

Collection

- A collection is a container that stores multiple documents inside a database.
- **Database** → contains collections
- **Collection** → contains documents (group of documents)
- **Document** → contains data (in JSON/BSON format)



CRUD

- C → Create
- R → Read
- U → Update
- D → Delete

To Create Database

use universityDB

- Switch to that database
- BUT it is not actually created.

- Create: Insert document into collection

```
db.students.insertOne({  
  name:(space) "Siraj",  
  age: (space) 20  
});
```

- Read : Fetch data from Collection

```
db.students.find({ name: "Bob" })
```

- Update : Modify existing Document

```
db.students.updateOne(  
  { name: "Bob" },  
  { $set: { age: 21 } }  
)
```

- Delete : Remove Document from Collection

db.students.deleteOne({ name: "Bob" })